REP-049 · Approvals, and PLAN-020 step 1: supply and customer demand measured

Game	Fruit Market (rules version 4 on the preview; 1.1.1 on the live game)
Report date	2026-10-27 23:20 (Asia/Kuala_Lumpur)
Type	Approval record and measurement report. **No game change**
Request	"do all except 22, for 22 analyse demand as well, feels like theres too much customer towards the final few turns"
Plans	PLAN-020, PLAN-021, PLAN-023 approved and in progress; PLAN-022 not approved

STATUS

PLAN-020 step 1 (measurement) done. Steps that change rules, speed or the server wait for your choice. The game files are unchanged (checked against git). Branch `workflow/decisions-020`, records only, not pushed.

SUMMARY

Both your impressions are confirmed in simulation: supply runs short and customers pile up late in the game. The biggest single cause is the Market Days with no lorry (days 3, 6, 9). Putting the lorry back on Market Days is the biggest single fix in the tests. Results come from computer players only.

PLAN OR WORK COMPLETED

How I read your message. "all except 22" = approve PLAN-020, 021 and 023, not 022. "for 22 analyse demand" = PLAN-020, which now also analyses demand. If that is wrong, say so.

Method. I installed the repo's dependencies from the lockfile in the local clone (no new dependency) and ran simulated games through the real engine: 300 games per setting for the baseline, 200 for each what-if, all computer shops at Normal, Standard 10 days with events, specialities and unripe fruit on. Per day, the script recorded the fruit on the market when each shop's buying turn starts, the customers, the fruit they wanted and the fruit sold. The what-ifs were temporary edits to a copy of the rules in the working folder, reverted afterwards; nothing was committed. Script and raw results are saved in the owner's Guides folder, `supply_demand_measurement`.

Baseline (today's rules, 10 days)

Shops	First 3 days: customers/day	Last 3 days: customers/day	Fill rate first / last	Shop turns short of their buy limit, first / last	Shop turns with an empty market, first / last
2	2.1	4.9	57% / 36%	4% / 32%	0% / 10%

Shops	First 3 days: customers/day	Last 3 days: customers/day	Fill rate first / last	Shop turns short of their buy limit, first / last	Shop turns with an empty market, first / last
3	3.5	6.9	52% / 32%	9% / 36%	3% / 17%
4	4.9	8.9	52% / 30%	12% / 35%	5% / 22%

(Fill rate = fruit sold divided by fruit the customers wanted.)

What this shows (4 shops)

- **Demand nearly doubles.** Customers per day go from 4.9 to 8.9; the fruit they want goes from about 19 to about 35.
- **Fruit sold per day stays flat** at about 14 from day 4 on. Supply does not grow with demand.
- **Market Days starve the market.** On days 3, 6 and 9 shops find an empty market on 15%, 41% and 49% of their turns (units on the market at a shop's turn start fall to 6.0, 3.5 and 2.7, against 13 to 18 on other days). Fill rate on day 9 is 12%.
- **Why demand grows:** one extra walk-in in the second half, one extra on each Market Day, and unserved customers coming back the next day. About 7 to 8 customers a day go unserved late in the game.
- **Quick (8 days, 4 shops):** the same pattern: last 3 days 8.7 customers a day, 33% fill, 33% of turns short, 21% empty.

What-ifs (4 shops, 10 days, last 3 days)

Variant	Customers/day	Fruit wanted/day	Fill rate	Turns short	Turns with an empty market	Empty on days 3 / 6 / 9
Today	8.9	35.5	30%	35%	22%	15% / 41% / 49%
A: lorry also comes on Market Days	7.7	28.4	59%	8%	2%	1% / 2% / 2%
D: +1 of each fruit in the second half	8.7	34.0	37%	25%	17%	15% / 41% / 39%
C: no extra walk-in in the second half	6.8	25.8	41%	31%	17%	15% / 37% / 38%
A + D	7.1	26.4	71%	2%	0%	1% / 2% / 0%
A + C	5.4	20.0	75%	2%	0%	1% / 2% / 0%

Reading the what-ifs

- **A alone fixes most of the supply problem** and also lowers late demand (7.7 vs 8.9), because fewer customers are left waiting. It is the simplest to play by hand: a farmer card every day.
- **D and C each help only a little** because they leave the Market Day starvation in place.
- **A + C** makes late-game demand level with early-game (5.4 vs 5.0 customers a day) and gives the highest fill rate; it removes one extra customer per day in the second half, a printed number.

- **Untested idea:** a half delivery on Market Days (keeps some scarcity). It needs an extra rule, so I did not test it.
- **Trade-off for A:** Market Days lose their "nothing arrives today" tension and keep only the extra walk-in and Last Call. That is a design call for you.

VALIDATION

Check	Result	Evidence
PLAN-020 T1 measurement report	PASS	This report; 2, 3 and 4 shops, 300 games each; what-ifs 200 games
PLAN-020 T2 owner target set	NOT RUN	Awaiting the owner
Game files unchanged	PASS	`git status` shows no change to src, server or RULES.md; temporary edits reverted (line endings only, restored with git)
Human evidence	NOT RUN	Computer players only; all Normal; no humans
`npm run verify`	NOT RUN	No game change to verify

CHANGELOG

Added: none for players. Changed: records only. State: unreleased.

RISKS AND COSTS

- **Computer players only.** They rarely undercut and may buy differently from people, so absolute numbers may not match human games. The direction is clear; the size is uncertain.
- **One configuration.** All Normal computers, events on. Easy or Hard computers, or the daily challenge and League settings, were not measured.
- Any change bumps the rules version and must be mirrored on the tabletop (DEC-027). A and C are simple printed rules; D needs a "from day 6" note on the table.
- Cost: RM0. The dependency install is on the C: drive, outside Drive.

NEXT ACTION OR APPROVAL

Decisions for you (PLAN-020 step 2):

1. Which option: A (recommended first), A + C, A + D, or something else such as a half delivery.
2. Your target before building, for example: "no more than 10% of shop turns find fewer fruit than their buy limit, in any 3-day stretch, with 4 shops". A meets it at 8%. Then step 3 (build, bump the rules version, update RULES.md and tests) can start.

PLAN-021 (session length): read-only findings so far, no build yet. Computer players wait 380 ms per action (150 ms with the existing Fast setting), which adds up to roughly 10 to 15 seconds a day with 3 computers; human thinking time is the bigger part. I propose: default length Quick for 3 to 4

shops, turn the existing fast setting on by default, then time a game with a stopwatch before and after. Needs your time target (proposed 20 minutes or less).

PLAN-023 (online undo and rename): design note ready for your approval (Design 1 recommended: server-side undo of the current turn; rename in the lobby only).

DOCUMENT RECORD

Report REP-049. Plans PLAN-020, PLAN-021 and PLAN-023 approved (evidence: "do all except 22, for 22 analyse demand as well, feels like theres too much customer towards the final few turns"), in progress; PLAN-022 proposed, not approved. PLAN-020 T1 PASS. Register: `docs/workflow/DECISIONS.md` . **PDF EXPORT BLOCKED:** pandoc is not installed on this machine (the repo's dependencies now are); recovery: install pandoc, then run `npm run records` (needs owner approval).